

Introduction to Business Cycle Data

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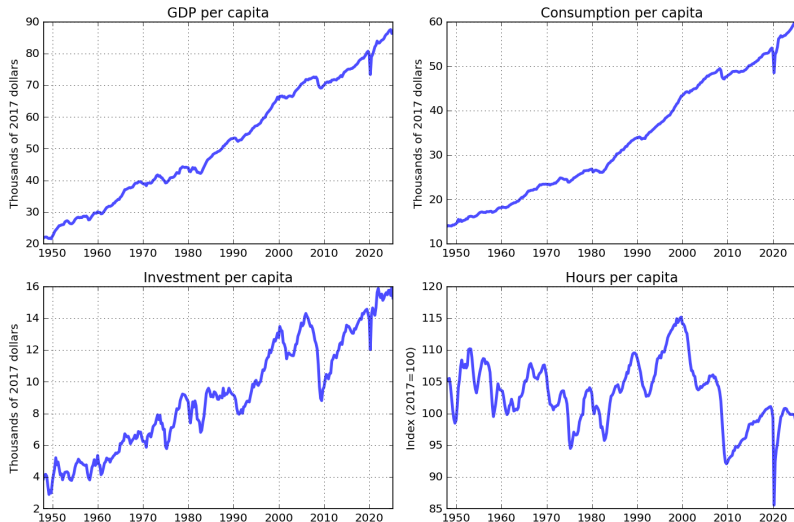
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- The *business cycle* is the fluctuation of many macroeconomic quantities that last for about 1.5 to 8 years.
- Business cycle fluctuations are costly:
 - Misallocations of capital and labor.
 - Particularly painful for workers that become unemployed and for the families of workers who become unemployed.

- We will examine two historically-competing schools of thought:
 - ① **Real Business Cycle (RBC)** theory: fluctuations in real quantities are primarily due to TFP shocks; i.e., shocks to the production function.
 - ② **New-Keynesian (NK)** theory: fluctuations are largely driven by aggregate demand and affect real quantities because of nominal rigidities (e.g., sticky prices).
- Both approaches have merits and shortcomings and elements of both are integrated into contemporary business cycle theory.

Figure 1: GDP, consumption, investment, and hours for the US from January 1948 to April 2025. Source: FRED.



Trend and Cycle Components

- Suppose that the value of a time series process X_t can be decomposed into two components: a *trend* component and a *cyclical* component.

$$X_t = X_t^{trend} + X_t^{cycle} \quad (1)$$

- The trend component is the long-run path about which the series fluctuates.
- The cyclical component is the difference between the value of a time series and the trend:

$$X_t^{cycle} = X_t - X_t^{trend} \quad (2)$$

Trend and Cycle Components

- Often, it's useful to express the cyclical component of a time series as the percent deviation of the series from trend (divided by 100):

$$\hat{x}_t = \frac{X_t - X_t^{trend}}{X_t^{trend}} = \frac{X_t^{cycle}}{X_t^{trend}} \quad (3)$$

- Note:

$$\frac{X_t - X_t^{trend}}{X_t^{trend}} = \frac{X_t^{cycle}}{X_t^{trend}} \approx \log X_t - \log X_t^{trend} \quad (4)$$

Trend and Cycle Components

Example: Percent Deviation from Trend

- Suppose:

$$X_t = 220 \quad (5)$$

$$X_t^{trend} = 215 \quad (6)$$

- Then:

$$\frac{X_t - X_t^{trend}}{X_t^{trend}} = \frac{220 - 215}{215} = 0.0233 \quad (7)$$

and:

$$\log X_t - \log X_t^{trend} = \log 220 - \log 215 = 0.0230 \quad (8)$$

Figure 2: GDP, consumption, investment, and hours per capita for the US from January 1948 to April 2025. Source: FRED.

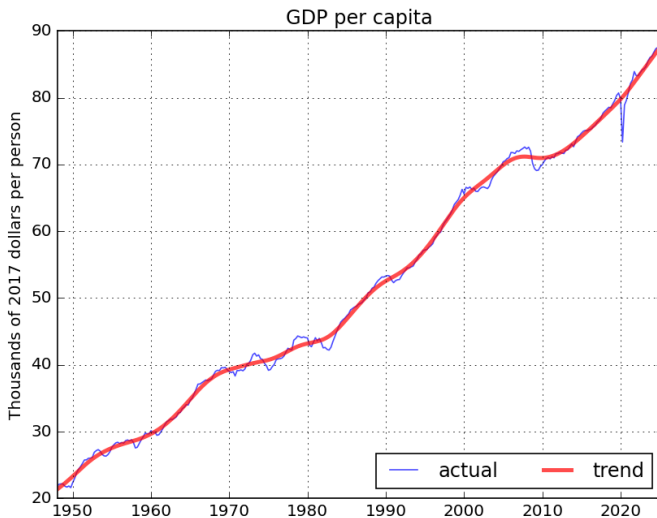


Figure 3: US GDP per capita: actual, trend, and cycle from January 1948 to April 2025. Source: FRED.

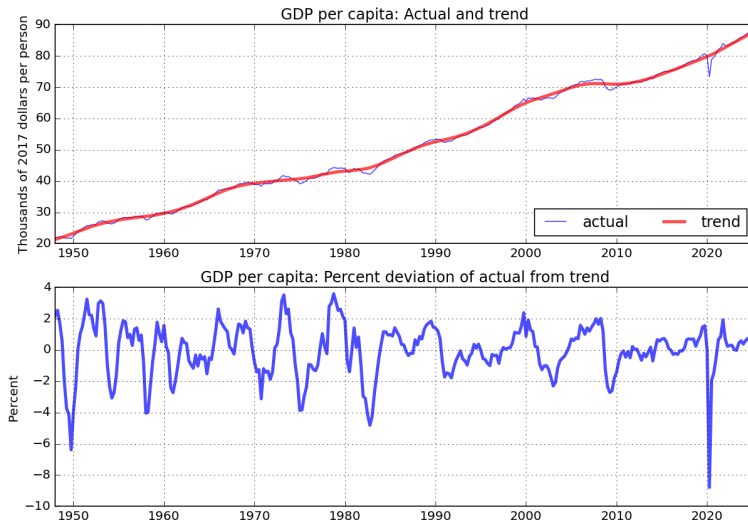


Figure 4: Business cycle components of GDP, consumption, investment, and hours for the US from January 1948 to April 2025.

Source: FRED.

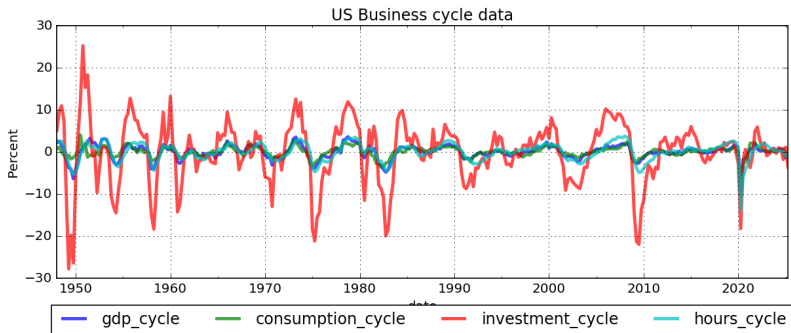


Table 1: Standard deviations of real business cycle data from January 1948 to April 2025. Units are percent deviations from trend.

Source: FRED.

GDP	1.649000
Consumption	1.310000
Investment	7.250000
Hours	2.081000

Table 2: Correlations of real business cycle data from January 1948 to April 2025. Units are percent deviations from trend. Source: FRED.

	GDP	Consumption	Investment	Hours
GDP	1.000000	0.810000	0.834000	0.869000
Consumption	0.810000	1.000000	0.642000	0.751000
Investment	0.834000	0.642000	1.000000	0.749000
Hours	0.869000	0.751000	0.749000	1.000000